

# AI Voice Agent Platform

Complete feature catalogue — a multi-tenant platform for automated and human-assisted calls on phone and WhatsApp

PHONE SIM / PSTN calls

WA-CALL WhatsApp calls

WA-MSG WhatsApp messages

ALL every channel

## 1 Accounts, workspaces and access

*Every company is an isolated tenant. Nothing is shared between them.*

|                             |                                                                                                                            |     |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Company sign-up</b>      | Self-serve registration with email and password, or Google. Email verification, password reset, and invitation acceptance. | ALL |
| <b>Workspaces</b>           | One company = one workspace, fully isolated. A user may belong to several and switch between them without signing out.     | ALL |
| <b>Roles</b>                | Owner, Admin, Operator and Viewer. Operators can take calls but cannot change credentials; Viewers read only.              | ALL |
| <b>Team invitations</b>     | Invite by email with a pre-assigned role. Pending invitations can be resent or revoked.                                    | ALL |
| <b>Audit trail</b>          | Every credential change, setting change, agent go-live and deletion recorded with actor, action, target and time.          | ALL |
| <b>Guided onboarding</b>    | A resumable six-step wizard: company profile, agent, knowledge base, channel choice, credentials, then test and go live.   | ALL |
| <b>White-label branding</b> | Company logo and accent colour theme the whole workspace. No rebuild, no per-customer deployment.                          | ALL |

## 2 The voice agent

*One agent definition drives every channel, so a company writes its persona once.*

|                                |                                                                                                                                             |                  |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Multiple agents</b>         | Any number of named agents per company, each in draft, live or paused, each with its own persona, voice and channels.                       | ALL              |
| <b>Persona builder</b>         | Structured form for agent name, role, greeting, language policy, tone, forbidden phrases, escalation rules and closing behaviour.           | ALL              |
| <b>Compiled prompt preview</b> | The system prompt is assembled live beside the form, so a company sees exactly what its agent was told.                                     | ALL              |
| <b>Prompt override</b>         | Power users can replace the compiled prompt wholesale, with a warning that it detaches from the form.                                       | ALL              |
| <b>Persona templates</b>       | Ready-made starting points: Admissions, Customer support, Collections, Appointment booking, Lead qualification.                             | ALL              |
| <b>Spoken greeting</b>         | A line the agent says verbatim to open every call, set per agent.                                                                           | PHONE<br>WA-CALL |
| <b>Gender-correct speech</b>   | The agent's gender drives verb agreement in languages that inflect by speaker, so it never breaks character mid-sentence.                   | PHONE<br>WA-CALL |
| <b>Language mirroring</b>      | Replies in whatever the caller just used - English, Urdu, Roman Urdu or a mix - and switches with them mid-call.                            | ALL              |
| <b>Voice selection</b>         | Voice, speaking speed and language per agent, with preview. Speed defaults below 1.0 because long vowels clip through a phone codec.        | PHONE<br>WA-CALL |
| <b>Barge-in</b>                | The caller can interrupt mid-sentence and the agent stops speaking immediately.                                                             | PHONE<br>WA-CALL |
| <b>Turn-taking controls</b>    | Voice-activity threshold, silence window, prefix padding, eagerness and turn-detection mode, with safe defaults most companies never touch. | PHONE<br>WA-CALL |

|                       |                                                                                                                  |                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Noise handling</b> | Noise reduction on the inbound leg, plus a transcription prompt that biases toward the company's own vocabulary. | PHONE<br>WA-CALL |
| <b>Call pacing</b>    | Maximum call duration, wrap-up warning before the limit, connect timeout and hang-up grace period.               | PHONE<br>WA-CALL |
| <b>Model choice</b>   | Speech model, reasoning effort and output limit per agent; a separate, cheaper text model for the chat feature.  | ALL              |

### 3 Knowledge base and retrieval

*Retrieval is given to the agent as a tool, not run as a pipeline stage, which is why answers stay near conversational speed.*

|                                 |                                                                                                                                                     |     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Document upload</b>          | Drag and drop PDF, DOCX, TXT and Markdown. Per-file progress, page count and chunk count.                                                           | ALL |
| <b>Structure-aware chunking</b> | Split on the document's own section markers rather than a blind token window, so a fee table stays one fact.                                        | ALL |
| <b>Table extraction</b>         | Tables are pulled row-wise so a programme, its per-credit rate and its total remain a single retrievable line.                                      | ALL |
| <b>Embedding index</b>          | Documents are embedded and indexed on upload, cached, and rebuilt only when the file or the settings change.                                        | ALL |
| <b>Grounded answers</b>         | The agent calls a knowledge search tool mid-conversation and speaks only what the documents support.                                                | ALL |
| <b>Refusal threshold</b>        | Below a tunable similarity score the agent says it does not know rather than inventing an answer. This is the anti-hallucination control.           | ALL |
| <b>Retrieval tuning</b>         | Chunk size, overlap, embedding model, top-K, chunks per section, context budget and the threshold, all editable per company.                        | ALL |
| <b>Test search console</b>      | Type a question, see ranked chunks with scores, with the threshold drawn across the results and a verdict on whether the agent would have answered. | ALL |
| <b>Per-document status</b>      | Each file reports parsed, indexed or failed, with the real error, and can be re-indexed or deleted on its own.                                      | ALL |
| <b>Text chat (Ask)</b>          | A written chat against the same knowledge base, with cited sources, section, page and score - for staff training and for checking answers.          | ALL |

### 4 Phone channel (SIM / PSTN)

*A real phone number. The carrier bridges the caller to the speech model directly, so audio never queues inside the platform.*

|                                    |                                                                                                                          |       |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Inbound answering</b>           | Calls to the company's number are answered by the agent automatically, around the clock.                                 | PHONE |
| <b>Outbound dialling</b>           | Place a call to any number from the dashboard; the agent opens with the configured greeting.                             | PHONE |
| <b>Bulk and scheduled outbound</b> | Queue a list of numbers for the agent to work through, with pacing and retry rules.                                      | PHONE |
| <b>Manual phone bridge</b>         | A staff member's own phone is dialled first, then the customer, and the two legs are bridged - no headset or app needed. | PHONE |
| <b>Browser softphone</b>           | Staff can take or place calls from the browser over WebRTC, with a short-lived access token issued per session.          | PHONE |
| <b>Mid-call takeover</b>           | A human can step onto a live agent call and continue it.                                                                 | PHONE |
| <b>Caller ID control</b>           | The company's own verified number is presented on outbound calls.                                                        | PHONE |
| <b>Number normalisation</b>        | Locally formatted numbers are expanded to international format before dialling, since carriers accept nothing else.      | PHONE |
| <b>Call reconciliation</b>         | A periodic sweep compares platform records against the carrier's, so a dropped webhook never leaves a call stuck open.   | PHONE |

## 5 WhatsApp calling channel

*Voice calls over WhatsApp Business Calling, with media terminated inside the platform so audio can be recorded or handed to a person.*

|                                 |                                                                                                                                               |                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Inbound WhatsApp calls</b>   | Incoming WhatsApp calls are accepted and answered by the agent, with optional pre-accept for faster audio.                                    | WA-CALL           |
| <b>Outbound WhatsApp calls</b>  | Call a WhatsApp user from the dashboard, with ring and answer timeouts surfaced in the UI.                                                    | WA-CALL           |
| <b>Call permission checks</b>   | Before dialling, the platform checks whether the user has permitted calls, and shows the reason when they have not.                           | WA-CALL           |
| <b>Permission requests</b>      | Send a permission request to a user who has not yet allowed calls, and track its status.                                                      | WA-CALL           |
| <b>Human counselor takeover</b> | A staff member answers the WhatsApp call themselves, speaking through the browser microphone instead of the agent.                            | WA-CALL           |
| <b>Two-way media bridge</b>     | WhatsApp audio is terminated in-platform and bridged to the speech model or to a person, which is what makes recording and takeover possible. | WA-CALL           |
| <b>NAT traversal</b>            | STUN and TURN configuration, so media connects from restrictive networks rather than failing after signalling succeeds.                       | WA-CALL           |
| <b>Agent self-test</b>          | Generate a sample of the agent speaking, without placing a call, to verify voice and persona before going live.                               | WA-CALL           |
| <b>Webhook signature checks</b> | Every incoming event is verified cryptographically; unsigned traffic is rejected outright.                                                    | WA-CALL<br>WA-MSG |

## 6 WhatsApp messaging channel

*Written follow-up, in the same thread as the call.*

|                               |                                                                                                                                 |        |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>Free-form messages</b>     | Send a WhatsApp message to any number from the dashboard, or in reply inside a 24-hour window.                                  | WA-MSG |
| <b>Template messages</b>      | Send approved templates with their variables filled, for business-initiated contact outside the reply window.                   | WA-MSG |
| <b>Template manager</b>       | List the company's templates with approval status, body preview and variable mapping, pulled from their own account.            | WA-MSG |
| <b>Approval-aware tooling</b> | An unapproved template means the agent is not offered that tool at all, so it never promises a message that cannot be sent.     | WA-MSG |
| <b>Agent-sent follow-up</b>   | Mid-call, the agent can offer to send details in writing and dispatch them itself once the caller agrees.                       | WA-MSG |
| <b>Reference tokens</b>       | An unresolved question becomes a short reference the caller can quote later, sent to them by message and tracked to resolution. | WA-MSG |
| <b>Inbound replies</b>        | When a customer replies in writing, the agent answers from the same knowledge base, in the same thread.                         | WA-MSG |
| <b>Length guards</b>          | Topic and detail length limits, so a spoken answer never becomes an unreadable wall of text.                                    | WA-MSG |
| <b>Delivery status</b>        | Sent, delivered, read and failed per message, with the provider's reason on failure.                                            | WA-MSG |
| <b>Per-contact threads</b>    | The full written history with one person in one view.                                                                           | WA-MSG |

## 7 Recording, transcripts and summaries

*What was said, kept and searchable.*

|                                |                                                                                                                                                  |                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Call recording</b>          | Record calls on either channel, with a per-company switch and a separate switch for the agent's own track.                                       | PHONE<br>WA-CALL |
| <b>Two-way capture</b>         | Both voices in one file. Each recording also declares its scope - both sides, caller only, or agent only - so nothing is silently half-captured. | PHONE<br>WA-CALL |
| <b>Independent recorder</b>    | Audio is captured by the platform rather than relying on the carrier, so a provider that drops a recording cannot lose the call.                 | PHONE<br>WA-CALL |
| <b>Honest recording states</b> | Recording, still composing, ready, or never recorded. A call with no audio says so plainly instead of offering a retry that cannot succeed.      | PHONE<br>WA-CALL |
| <b>Playback</b>                | In-browser player with waveform, speed control, seeking from the transcript, and download.                                                       | PHONE<br>WA-CALL |
| <b>Transcoding</b>             | Recordings are converted to a compact, universally playable format for storage and sharing.                                                      | PHONE<br>WA-CALL |
| <b>Durable archive</b>         | Recordings are copied to object storage with expiring signed links, so they survive a host with no persistent disk.                              | PHONE<br>WA-CALL |
| <b>Live transcripts</b>        | Speech is transcribed as the call happens, speaker-labelled and timestamped, in Urdu or English.                                                 | PHONE<br>WA-CALL |
| <b>Stored transcripts</b>      | The full transcript is kept, searchable, and linked to playback position.                                                                        | PHONE<br>WA-CALL |
| <b>Post-call summary</b>       | An automatic summary of what was asked, what was answered and what happens next, generated when the call ends.                                   | PHONE<br>WA-CALL |
| <b>Outcome classification</b>  | Each call is tagged with how it ended, for filtering and reporting.                                                                              | ALL              |
| <b>Retention policy</b>        | How long recordings and transcripts are kept, set per company.                                                                                   | ALL              |

## 8 Live operations

*What is happening right now, for a supervisor watching the floor.*

|                                |                                                                                                                                                             |                  |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Live call monitor</b>       | Every active call, with caller, channel, direction, agent and elapsed time, updating in real time.                                                          | ALL              |
| <b>Streaming transcript</b>    | The conversation appears as it is spoken, auto-scrolling.                                                                                                   | PHONE<br>WA-CALL |
| <b>Speaking indicator</b>      | An animated voice orb showing who currently holds the floor.                                                                                                | PHONE<br>WA-CALL |
| <b>Retrieval activity feed</b> | Every knowledge-base lookup during the call, with its score against the threshold and whether it was answered - the fastest way to diagnose a wrong answer. | ALL              |
| <b>Listen in</b>               | Monitor a live call without joining it.                                                                                                                     | PHONE<br>WA-CALL |
| <b>Take over</b>               | Step onto a live call and continue it as a human.                                                                                                           | PHONE<br>WA-CALL |
| <b>Force end</b>               | End a call from the dashboard.                                                                                                                              | ALL              |
| <b>Connection state</b>        | A persistent indicator showing live, reconnecting or offline, so the floor never mistakes a stale screen for a quiet day.                                   | ALL              |
| <b>Operational dashboard</b>   | Calls today, talk time, answer rate, active calls, messages sent, a 14-day volume chart split by channel, and channel health at a glance.                   | ALL              |

## 9 History, search and reporting

*The record, and what companies do with it.*

|                              |                                                                                                                      |     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|-----|
| <b>Call history</b>          | Every call with time, direction, number, channel, who handled it, duration, outcome and recording state.             | ALL |
| <b>Filtering</b>             | By date range, channel, direction, agent or human, outcome and agent. Saved views for the filters a team uses daily. | ALL |
| <b>Transcript search</b>     | Find calls by what was said, not only by number or date.                                                             | ALL |
| <b>Call detail</b>           | Summary, transcript, recording and the raw event timeline for one call in one sheet.                                 | ALL |
| <b>Spreadsheet export</b>    | A multi-sheet workbook of calls, transcripts and summaries, generated on demand.                                     | ALL |
| <b>CSV export</b>            | The current filtered view, exported as it appears.                                                                   | ALL |
| <b>Drive and Sheets sync</b> | Optionally keep a live copy of the workbook in the company's own Google Drive.                                       | ALL |
| <b>Outbound webhooks</b>     | Push call and message events to the company's own systems, with selectable event types and a signing secret.         | ALL |
| <b>API keys</b>              | Scoped keys for a company's own integrations, shown as last-four only, with last-used time.                          | ALL |

## 10 Setup, credentials and self-serve guides

*Companies bring their own provider accounts, so the platform never resells minutes and never holds the customer relationship.*

|                                   |                                                                                                                                                                                                                                                      |     |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Bring your own credentials</b> | Each company connects its own telephony, WhatsApp and model accounts. Usage bills to them directly.                                                                                                                                                  | ALL |
| <b>Write-only secrets</b>         | A saved credential is never shown again - only its last four characters, who set it and when, with a Replace action.                                                                                                                                 | ALL |
| <b>Channel health</b>             | Each channel reports disconnected, pending, verifying, connected or error, with the provider's own error text and a link to the fix.                                                                                                                 | ALL |
| <b>Connection tests</b>           | Verify credentials before going live, rather than discovering the problem on a customer's call.                                                                                                                                                      | ALL |
| <b>Everything off by default</b>  | Every capability switch starts off, so saving credentials alone can never make an agent answer or dial.                                                                                                                                              | ALL |
| <b>Readiness gating</b>           | Any control that cannot yet succeed is disabled and says what is missing, with a link to the relevant guide.                                                                                                                                         | ALL |
| <b>Webhook helper</b>             | The exact callback URL and verification token to paste into the provider's dashboard, with copy buttons.                                                                                                                                             | ALL |
| <b>Inline walkthroughs</b>        | Beside every credential field, where to find that value in the provider's own dashboard.                                                                                                                                                             | ALL |
| <b>Guides library</b>             | Twelve written guides covering first setup, provider credentials, number verification, webhooks, writing a knowledge base, tuning the threshold, persona writing, template approval, recording and consent, error codes, and a pre-launch checklist. | ALL |
| <b>Contextual help</b>            | Empty states and disabled controls link to the guide that unblocks them, so help arrives where the work stops.                                                                                                                                       | ALL |

## 11 Usage, limits and billing

|                           |                                                                                       |     |
|---------------------------|---------------------------------------------------------------------------------------|-----|
| <b>Usage metering</b>     | Call minutes, messages and storage metered per company and per channel.               | ALL |
| <b>Plan limits</b>        | Minutes, messages, storage, agents and seats, each shown as a meter against the plan. | ALL |
| <b>Limit warnings</b>     | Visible warnings at 80 and 100 per cent of any limit, before service is affected.     | ALL |
| <b>Cost breakdown</b>     | Spend split by channel, so a company can see what voice costs against messaging.      | ALL |
| <b>Plans and upgrades</b> | Tier comparison with in-product upgrade, and invoice history.                         | ALL |

## 12 Platform quality

*Cross-cutting behaviour that applies to every screen.*

|                              |                                                                                                                                    |     |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Bilingual interface</b>   | The whole product in English and Urdu, with correct Nastaliq typesetting and full right-to-left layout.                            | ALL |
| <b>Light and dark themes</b> | Both designed first-class, not an inversion, and both respecting the system preference.                                            | ALL |
| <b>Accessibility</b>         | Full keyboard navigation, visible focus, screen-reader labelling on live regions and meters, and verified contrast in both themes. | ALL |
| <b>Mobile and tablet</b>     | Every screen usable at phone width; supervisors can watch the floor from a handset.                                                | ALL |
| <b>Honest states</b>         | Designed empty states, skeleton loaders shaped like real content, and errors that say what failed and what to try.                 | ALL |
| <b>Safe destruction</b>      | Every destructive action confirms and names exactly what will be lost.                                                             | ALL |
| <b>Command palette</b>       | Keyboard-driven navigation and actions from anywhere in the product.                                                               | ALL |

**Delivery note.** The current build covers the full interface and experience for everything above, running against realistic in-product sample data. Connecting it to the live calling, messaging and retrieval services is the following phase: those services already exist and run today as two single-company systems, and the work is to give them a company dimension and a shared settings store. The interface is deliberately built against a single typed data layer so that connection happens in one place.